Intelligent Reading Comprehension and Quiz Generation System

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1. Abstract

This project presents an end-to-end intelligent machine learning pipeline for automated reading comprehension question generation and verification using the RACE dataset. The system comprises two core components: Model A, which performs answer verification by predicting the correct option among four candidates, and Model B, which generates plausible distractors and graduated hints to support active learning. For Model A, multiple traditional classifiers were trained and evaluated. Random Forest achieved the best macro F1 score of 0.505, while Label Propagation (semi-supervised) achieved the highest accuracy at 72.30% using only 20% labeled data, significantly outperforming fully supervised baselines. For Model B, a Random Forest distractor ranker achieved 98.61% accuracy and a macro F1 of 0.9303, with an extractive hint classifier achieving 98.27% accuracy and perfect recall of 1.000. The complete pipeline was deployed as a four-screen Streamlit web application integrating article input, interactive quiz, graduated hint panel, and an analytics dashboard. This work demonstrates the practical viability of classical machine learning techniques in educational technology while acknowledging key ethical constraints around automated assessment, dataset bias, and model transparency.

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2. Introduction & Motivation

Reading comprehension is a foundational skill in language education. However, the manual creation of high-quality reading comprehension questions, plausible distractors, and contextual hints is an extremely time-consuming and labour-intensive process for educators. The primary motivation of this project is to automate and assist this educational workflow using classical Machine Learning techniques applied to a large-scale benchmark dataset.

The system is designed around two objectives. First, it must verify whether a candidate answer is correct given a passage and a question — a classification task requiring the model to understand textual entailment between the article, question, and option text. Second, it must generate its own distractors and supportive hints to guide the student toward the correct answer without directly revealing it.

By automating these tasks, educators can scale the generation of practice materials significantly. Students simultaneously receive immediate, guided feedback through graduated hints that promote active learning rather than passive answer reveal. This project demonstrates how traditional machine learning, when combined with thoughtful feature engineering, structured lexical resources, and a user-centric design, can produce a robust and functional educational tool.

The system is built entirely on classical ML techniques — including Logistic Regression, Support Vector Machines, Random Forest, K-Means, Gaussian Mixture Models, and Label Propagation — without reliance on deep neural networks, making it computationally efficient and deployable on standard university hardware.

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3. Related Work

The field of automated question generation and reading comprehension has been shaped by several landmark contributions spanning both classical and neural approaches.

1. **Lai et al. (2017)** introduced the RACE dataset (ReAding Comprehension from Examinations), compiled from real Chinese middle and high school English examinations. RACE is notable for its scale (approximately 28,000 passages and 100,000 questions) and its emphasis on multi-hop reasoning rather than simple fact retrieval, making it substantially harder than prior benchmarks such as SQuAD.
2. **Devlin et al. (2019)** proposed BERT (Bidirectional Encoder Representations from Transformers), which established a new state of the art across NLP benchmarks through deep bidirectional pretraining. While our project focuses on traditional ML baselines, BERT represents the performance ceiling for semantic reasoning tasks on RACE and provides context for interpreting our results.
3. **Guo et al. (2016)** investigated automated distractor generation for reading comprehension questions, establishing that effective distractors must simultaneously satisfy four properties: plausibility to an uninformed reader, factual incorrectness with respect to the passage, diversity across the three options, and grammatical consistency with the correct answer. These four constraints directly informed the design of our Model B pipeline.
4. **Du, Shao, and Cardie (2017)** demonstrated that neural sequence-to-sequence models could generate fluent, relevant questions directly from text segments, outperforming earlier rule-based and template-driven approaches. Their work motivates the template-plus-ranker

generation strategy we adopted for Model A's generation sub-task under classical ML constraints.

5. **Papineni et al. (2002)** introduced BLEU (Bilingual Evaluation Understudy), and **Lin (2004)** introduced ROUGE (Recall-Oriented Understudy for Gisting Evaluation). Both metrics remain the standard for evaluating the quality of generated text against human references, and we apply them to evaluate our hint and distractor generation outputs in Model B.

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4. Dataset Analysis

4.1 Dataset Overview

The RACE dataset was used exclusively throughout this project. It consists of English reading passages with associated multiple-choice questions, each having four candidate options (A, B, C, D) and a single correct answer label.

Metric	Value
Total Passages	28,000+
Total Questions	~100,000
Question Types	Multiple-choice (A / B / C / D)
Source	Chinese middle & high school English exams
Language	English

4.2 Data Split

The dataset was divided into three non-overlapping splits for training, validation, and testing:

Split	Passages	Questions (approx.)
Train	17,594	87,866
Validation	3,770	~18,000
Test	3,771	~18,000

4.3 Column Schema

Each row in the dataset contains the following fields:

Column	Type	Description
`id`	String	Unique identifier
`article`	String	Full reading passage
`question`	String	Multiple-choice question
`A`	String	Answer option A
`B`	String	Answer option B
`C`	String	Answer option C
`D`	String	Answer option D
`answer`	String	Correct answer label (A, B, C, or D)

4.4 Exploratory Data Analysis

Initial EDA revealed significant variation in passage length, with middle-school passages tending to be shorter and more narrative, while high-school passages were longer and more argumentative. Question types included detail retrieval, inference, vocabulary-in-context, and main-idea questions. No missing values were found across any split.

4.5 Class Imbalance

The binary formulation of Model A's verification task — predicting whether a given option is correct or incorrect — inherently produces a 3:1 class imbalance, since each question has three incorrect options for every one correct option. This imbalance was addressed directly during model training by applying balanced class weighting to penalize minority-class misclassifications more heavily.

4.6 Preprocessing Pipeline

The following normalization steps were applied consistently across all splits:

- Lowercasing of all text
- Punctuation removal
- Whitespace collapsing
- Stop word retention (intentional — words such as "not" and "except" carry critical logical cues for reading comprehension)
- Concatenation of article, question, and option text into a single feature string per training example

Feature representation used TF-IDF vectorization with a 50,000-word vocabulary, producing sparse matrices suitable for all classical ML classifiers used in this project. One-Hot Encoding was used as an additional feature representation for distractor and hint generation in Model B.

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5. Model A: Design, Training, Results

5.1 Problem Definition

Model A performs two sub-tasks:

- **Verification:** Given an article, question, and candidate option, predict whether the option is the correct answer (binary classification).
- **Generation:** Given an article and the correct answer, produce a meaningful multiple-choice question using a template-based pipeline ranked by a trained ML classifier.

5.2 Feature Engineering

TF-IDF vectorization (50,000-word vocabulary, scikit-learn) was applied to the concatenated string of article + question + option. Additional handcrafted lexical features included keyword overlap scores, answer span presence in the article, and option length ratios.

5.3 Supervised Learning Models

At least two traditional ML classifiers were trained and compared on the verification sub-task. All models were evaluated on the held-out test set.

Model	Accuracy	Precision (Macro)	Recall (Macro)	F1 (Macro)
Logistic Regression	53.29%	0.535	0.545	0.523
Linear SVC (Calibrated)	74.99%	0.589	0.500	0.429
Naive Bayes	75.00%	0.375	0.500	0.428
Random Forest	**64.72%**	**0.542**	**0.546**	**0.505**
Soft-Vote Ensemble	74.41%	0.544	0.504	0.448

Key observations:

- Naive Bayes and Linear SVC achieved high accuracy figures (75%) but this is misleading — both models predominantly predict the majority class ("incorrect"), resulting in near-zero recall for the correct-answer class and a macro F1 below 0.43. Naive Bayes recorded a precision of 0.0 for the correct class, confirming complete failure to identify correct answers.
- Random Forest, despite a lower overall accuracy of 64.72%, achieved the highest macro F1 of 0.505 and the most balanced precision/recall trade-off (0.542 / 0.546), making it the most reliable model for the verification task.
- Logistic Regression achieved the most balanced recall (0.545) and is preferred when maximising detection of correct answers is prioritised over raw accuracy.
- The Soft-Vote Ensemble (combining LR, SVC, and RF probability outputs at threshold 0.49) achieved 74.41% accuracy but its macro F1 of 0.448 indicates it is still biased toward the majority class.

Random Forest was selected as the primary Model A classifier based on its superior macro F1 and balanced class performance.

5.4 Template-Based Question Generation

For the generation sub-task, a template-based pipeline was implemented:

1. **Step 1:** Candidate sentences extracted from the passage using keyword overlap scoring between sentence tokens and the correct answer, combined with position and length scores.
2. **Step 2:** Wh-word templates (Who / What / Where / When / Why / How) applied to candidate sentences based on answer type detection.
3. **Step 3:** Generated candidate questions ranked using a trained SVM/Random Forest classifier scoring fluency and relevance features. The highest-scoring question is returned.

5.5 Unsupervised and Semi-Supervised Approaches

In addition to supervised baselines, unsupervised and semi-supervised techniques were explored on dimensionality-reduced TF-IDF features (TruncatedSVD).

K-Means Clustering:
Grouping question-answer pairs into clusters revealed poor natural separation in TF-IDF feature space, with a silhouette score of 0.035 and a cluster correct-answer ratio of approximately 0.253–0.272. This confirms that sparse bag-of-words representations do not encode the semantic information needed to naturally cluster correct from incorrect answers.

Gaussian Mixture Models (GMM):
GMM probabilistic clustering achieved a silhouette score of 0.058 and a mean soft-label confidence of 0.962, indicating high-confidence but poorly calibrated assignments — the model was confident in cluster membership that did not correlate well with true labels.

Label Propagation (Semi-Supervised):

Metric	Value
Accuracy	72.30%
Precision (Weighted)	0.634
Recall (Weighted)	0.723
F1 (Weighted)	0.652
Labeled Data Used	20%

Label Propagation is the standout result of Model A. Using only 20% labeled data and propagating labels through the feature-space graph to 80% unlabeled samples, it significantly outperformed all fully supervised baselines in terms of accuracy. This finding suggests that reading comprehension answer spaces possess an underlying manifold structure that graph-based propagation methods can exploit effectively, making Label Propagation particularly valuable in low-label educational data scenarios.

5.6 Ensemble Strategy

A soft-vote ensemble combined probability outputs from Logistic Regression, Linear SVC (calibrated), and Random Forest. A decision threshold of 0.49 was tuned on the validation set to improve recall on the correct-answer class. The ensemble achieved 74.41% accuracy, though its macro F1 of 0.448 remains below Random Forest alone, demonstrating that ensembling does not always outperform the best individual model when class imbalance is severe.

5.7 Model A Evaluation Metrics Summary

Metric	Application
Accuracy	Fraction of correctly labeled options
Macro F1	Handles class imbalance — primary metric

Metric	Application
Exact Match (EM)	Strict character-level match to gold label
Silhouette Score	Clustering quality for K-Means and GMM
Weighted F1	Semi-supervised evaluation

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6. Model B: Design, Training, Results

6.1 Problem Definition

Model B takes a reading passage, question, and correct answer as input and produces:

- **Three plausible distractors:** Semantically related but factually incorrect answer options.
- **Three graduated hints:** Extractively scored passage sentences that guide the reader toward the correct answer without revealing it directly.

6.2 Distractor Generation Pipeline

Candidate Extraction:
Candidate phrases were extracted from the passage using three complementary sources:

1. **One-Hot Encoding cosine similarity:** Passage tokens ranked by cosine similarity to the correct answer token vector, retrieving lexically related candidates grounded in the passage.
2. **Word2Vec Nearest Neighbours (Gensim, Google News 300-d):** The correct answer was embedded using pre-trained Word2Vec vectors. The top-N most similar words not appearing in the passage were retrieved as semantically plausible external candidates.
3. **Frequency-Based Substitution:** High-frequency content words in the passage identified via frequency counting and substituted to produce additional candidates.

Feature Engineering per Candidate:

Feature	Description
`cosine_sim_to_answer`	One-Hot cosine similarity to correct answer
`w2v_similarity`	Word2Vec cosine similarity to correct answer
`char_length_diff`	Absolute character length difference from answer

Feature	Description
`freq_in_article`	Frequency of candidate token in passage
`is_in_question`	Binary flag — candidate appears in question text
`source_priority`	Integer encoding candidate source quality

ML Ranker Training:
A Random Forest and Logistic Regression ranker were trained on these features. The top-3 non-answer candidates by predicted score were selected as distractors. A diversity penalty was applied: if two candidates share more than 50% character bigrams, the lower-scored candidate's score is multiplied by 0.5 to enforce variation.

Distractor Ranker Results:

Model	Accuracy	Precision	Recall	F1
Logistic Regression	98.25%	0.9752	0.8479	0.9071
Random Forest	**98.61%**	**0.9417**	**0.9192**	**0.9303**

Distractor Generation Evaluation (BLEU / ROUGE vs. gold wrong options):

Metric	Score
BLEU Score	0.0038
ROUGE-1	0.0221
ROUGE-2	0.0014
ROUGE-L	0.0221
METEOR Score	0.0140
Ranker Precision	0.9989
Ranker Recall	0.1890
Ranker F1	0.3179
Ranker Accuracy	0.9638

The low BLEU and ROUGE scores for distractor generation reflect a known property of this task: good distractors are semantically related but lexically distinct from the gold reference distractors, meaning surface-level n-gram overlap metrics systematically underestimate distractor quality. The ranker precision of 0.9989 confirms that the model almost never selects the correct answer as a distractor, which is the primary correctness constraint.

6.3 Hint Generation Pipeline

Design:
An extractive hint scorer evaluates every sentence in the passage for relevance to the given question. Features used per sentence:

Feature	Description
`keyword_overlap`	Token overlap between sentence and question
`answer_overlap`	Token overlap between sentence and correct answer
`position_score`	Normalized sentence position in passage
`sentence_length_norm`	Normalized sentence word count
`contains_answer`	Binary — correct answer appears in sentence

A Logistic Regression classifier was trained on these features. The top-3 sentences by relevance score are surfaced as graduated hints:

- **Hint 1:** Most general — answer masked, broad context only
- **Hint 2:** More specific — answer partially masked
- **Hint 3:** Near-explicit — top-scoring sentence with close paraphrase of answer region

Hint Classifier Results:

Model	Accuracy	Precision	Recall	F1
Logistic Regression	98.27%	0.9218	1.0000	0.9593

Hint Generation Evaluation:

Metric	Score
BLEU Score	0.2362
ROUGE-1	0.3652

Metric	Score
ROUGE-2	0.2484
ROUGE-L	0.3315
METEOR Score	0.2986
Precision@3	0.8917
Classifier F1	0.9593

Hint generation achieved substantially higher BLEU (0.2362) and ROUGE-1 (0.3652) scores than distractor generation, reflecting that extractive hints share significant lexical overlap with gold answer-supporting sentences. A Precision@3 of 0.8917 means that in nearly 89% of cases, at least one of the three surfaced hint sentences directly overlaps with the gold answer-supporting sentence.

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7. User Interface Description

The system is deployed as a Streamlit web application providing four distinct screens integrated into a single coherent interface.

Screen 1 — Article Input

A full-width text area allows users to paste any reading passage. A "Load Random RACE Sample" button populates the text area with a random passage from the RACE test set for quick demonstration. A "Generate Quiz" button triggers both Model A and Model B inference simultaneously, with loading spinners displayed during processing to indicate system activity.

Screen 2 — Quiz View

Displays the generated or RACE-original question alongside four radio-button options (A, B, C, D) — one correct answer and three distractors generated by Model B. The user selects an option and clicks "Check Answer." Model A verification is applied and the result is displayed with colour-coded feedback: green for correct, red for incorrect. A supporting explanation highlights the relevant passage sentence.

Screen 3 — Hint Panel

Three progressive hints from Model B are presented using collapsible expander components. Hints are revealed sequentially — the user must open each hint before the next becomes available, preventing skipping. The "Reveal Answer" button appears only after all three hints have been expanded, enforcing the graduated learning flow.

Screen 4 — Analytics Dashboard

A tabbed dashboard presents:

- **Model A metrics:** Accuracy, macro F1, precision, recall, confusion matrix, and inference latency for the last N verification calls.
- **Model B metrics:** Distractor ranker accuracy, precision, recall, F1, and hint scorer precision@3 and classifier F1.
- **Session log:** A table of all questions attempted in the current session, exportable to CSV via a download button.
- **Latency tracking:** Per-request inference time displayed as a bar chart across session history.

UX Requirements Met

- The application is operable without reading documentation.
- All error states (empty input, model loading failure) display user-friendly messages.
- Loading indicators are shown during all model inference calls.
- Sufficient colour contrast and readable font sizes are maintained throughout.
- AI-generated content is labelled explicitly to maintain transparency with users.

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8. Evaluation & Discussion

8.1 Model A Discussion

The most important finding in Model A is the distinction between accuracy and macro F1 as evaluation metrics under class imbalance. Naive Bayes and Linear SVC both achieved approximately 75% accuracy by predominantly predicting the majority class ("incorrect"), yet their macro F1 scores of 0.428 and 0.429 respectively reveal near-complete failure to identify correct answers. This deceptive accuracy pattern is a direct consequence of the 3:1 class imbalance inherent in the RACE formulation.

Random Forest, by contrast, achieved the highest macro F1 of 0.505 with balanced precision and recall of 0.542 and 0.546 respectively. This confirms Random Forest as the most reliable traditional classifier for this task and underscores macro F1 as the appropriate primary metric when class distributions are skewed.

The Label Propagation result of 72.30% accuracy with only 20% labeled data is the most significant finding of Model A. It demonstrates that the feature space of TF-IDF-encoded reading comprehension examples contains exploitable manifold structure — nearby examples tend to share the same correct/incorrect label, allowing graph-based propagation to generalize effectively from a small labeled seed. This is practically valuable in educational settings where labeled data is scarce and expensive to produce.

The poor silhouette scores for K-Means (0.035) and GMM (0.058) confirm that unsupervised clustering on TF-IDF features cannot naturally separate correct from incorrect answers. The semantic signal required for this distinction is largely absent from bag-of-words representations.

8.2 Model B Discussion

The distractor ranker's high accuracy (98.61%) and precision (0.9989) confirm that the model almost never selects the correct answer as a distractor — the primary functional requirement. The lower recall of 0.1890 and F1 of 0.3179 on the generation evaluation reflect the difficulty of exactly recovering gold reference distractors from a lexical candidate pool, rather than a failure of distractor plausibility.

The low BLEU and ROUGE scores for distractor generation (BLEU: 0.0038, ROUGE-1: 0.0221) are expected and do not indicate poor distractor quality. Multiple plausible distractors exist for any given question-answer pair, and the gold reference distractors represent only one of many valid solutions. N-gram overlap metrics are poorly suited to evaluating distractor generation and are reported here for completeness only.

The hint generation results are stronger across all metrics (BLEU: 0.2362, ROUGE-1: 0.3652, Precision@3: 0.8917), reflecting that extractive hint selection from the passage has a more constrained solution space — the gold answer-supporting sentence is a single identifiable target, and the model recovers it in nearly 89% of cases within the top 3 ranked sentences.

8.3 Overall System Assessment

The combined system demonstrates that classical ML, carefully applied with structured feature engineering and multiple candidate sources, produces a functional educational quiz generation tool. The integration of Word2Vec alongside one-hot cosine similarity for distractor candidate retrieval meaningfully extended the candidate pool beyond passage-grounded tokens, producing distractors that are semantically plausible rather than merely lexically frequent.

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9. Limitations & Future Work

9.1 Technical Limitations

- **Model A ceiling:** The macro F1 ceiling of approximately 0.505 with traditional ML reflects the fundamental limitation of bag-of-words representations for reading comprehension. TF-IDF treats words as isolated tokens, losing word order, coreference, and multi-hop reasoning chains. The gap between our best traditional result and BERT-based models (which exceed 85% accuracy on RACE) quantifies this limitation directly.
- **Distractor lexical quality:** The candidate pool for distractor generation is bounded by passage vocabulary and Word2Vec coverage. Proper nouns, named entities, and domain-specific terms with no Word2Vec embedding fall through to the one-hot cosine fallback, which produces lower-quality candidates. WordNet integration would further extend categorical candidate coverage for common noun answers.
- **Template question generation:** The template-based question generation pipeline produces grammatically acceptable but often unnatural questions. The RACE dataset's questions frequently involve multi-sentence reasoning that cannot be reduced to a single Wh-word template applied to a candidate sentence.
- **Inference latency:** Latency is calculated dynamically at runtime in the Streamlit analytics dashboard. Future work should standardise latency benchmarking across a fixed test set to enable reproducible comparison.

9.2 Ethical Considerations

- **Dataset bias:** The RACE dataset originates exclusively from Chinese middle and high school English examinations. The vocabulary, cultural references, and question styles reflect a specific ESL curriculum in East Asia. Models trained on this data may generalise poorly to passages from other cultural contexts, dialects, or educational traditions.
- **Accessibility:** The current UI meets basic readability standards. Future iterations must strictly adhere to WCAG AA guidelines, including ARIA labels, full keyboard navigation, and screen reader compatibility for visually impaired users.
- **Academic integrity:** Generated questions and distractors must never be deployed in live examination settings without rigorous human review. Automated systems can produce subtly ambiguous or factually incorrect distractors that penalise knowledgeable students. The UI makes this constraint explicit to all users.
- **Model transparency:** The system explicitly labels all AI-generated content within the UI, ensuring users understand that errors are possible and that the system is a study aid rather than an authoritative assessment tool.

9.3 Future Work

- Replacing TF-IDF with contextual embeddings (BERT, RoBERTa) for Model A verification to capture multi-hop reasoning and coreference.
- Integrating WordNet co-hyponym lookup as a third distractor candidate source to improve categorical distractor quality.
- Fine-tuning T5-small on RACE for abstractive question generation to replace the current template-based approach.
- Extending evaluation with human Likert-scale ratings (1–5) for distractor plausibility and hint utility.
- Implementing WCAG AA compliance across the full Streamlit interface.

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10. Conclusion

This project successfully implemented an end-to-end classical machine learning pipeline for intelligent reading comprehension quiz generation using the RACE dataset. Model A demonstrated that Random Forest with balanced class weighting and macro F1 optimisation produces the most reliable answer verification classifier among traditional ML approaches, achieving a macro F1 of 0.505. The Label Propagation semi-supervised approach achieved the highest accuracy of 72.30% using only 20% labeled data, representing the most significant single finding of the project and highlighting the value of graph-based propagation for low-resource educational NLP tasks.

Model B produced a high-precision distractor ranker (98.61% accuracy, 0.9989 precision) and a high-recall extractive hint classifier (perfect recall of 1.000, Precision@3 of 0.8917), demonstrating that structured lexical feature engineering combined with Word2Vec semantic similarity can produce functionally useful educational support structures within classical ML constraints.

The complete system was deployed as a four-screen Streamlit application integrating both models into a coherent, accessible, and transparent user experience. While the performance gap relative to transformer-based systems remains substantial, this project establishes strong classical baselines, provides interpretable models suitable for educational deployment, and demonstrates the practical value of traditional machine learning in educational technology when applied with rigorous feature engineering and evaluation discipline.

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